Project-Proposal

Jaylen

Dataset 1: Screen Time, Sleep & Stress Analysis

Kaggle - Data was deleted from Kaggle, code below has been commented out

> This data was originally collected by Amar Tiwari and is synthetically generated for research, educational, and machine learning experimentation purposes. The data isn't representative of real individuals, however values were generated within realistic behavioral ranges to simulate real-world smartphone usage and productivity patterns
>
> Who: the subjects of the study are simulated people who use smartphones
>
> When: the data was created synthetically
>
> Where: the data was created synthetically
>
> Observations: 50000
>
> Variables: 13 (12 excluding user ID)
>
> Summary: screen time, sleep, stress, gender, age, etc.

Research Questions

1. To what extent does screen time impact stress, productivity, sleep and potentially caffeine intake?
   - Explanatory Variables: Daily phone hours, weekend screen time hours, app usage count, social media hours
   - Response Variables: Sleep hours, work productivity score, caffeine intake, stress level
2. Does the type of device impact a users screen time and sleep?
   - Explanatory Variables: Device type

- Response Variables: Daily phone hours, weekend screen time hours, sleep hours

Dataset 2: Resume Data

Kaggle

This dataset, curated and processed by Neuralframe AI. The author is Saugata Roy Arghya

Who: people who have submitted their resumes to companies or have posted them online publicly

- There was alleged explicit consent from the contributors

When: the data was created synthetically

Where: the data was curated by Neuralframe AI using both open-source data and proprietary company data

Observations: 9544

Variables: 35

Summary: includes information on career objectives, skills, education, work experience, certifications, and other relevant details pertaining to a persons resume

Research Questions

1. What information on a person's resume contributed the most to their match score for a specific job?
    - Explanatory Variables: Skills, institution, degree names, major(s), positions, etc.
    - Response Variables: Matched score
2. What is the average experience of candidates that have a job in their major?
    - Explanatory Variables: Major field of study, positions, skills, experience requirement, skills required
    - Response Variables: Role positions, job position name

```
Rows: 9,544
Columns: 35
$ address                                <chr> "", "", "", "", "", "", "", "", ""~
$ career_objective                       <chr> "Big data analytics working and da~
$ skills                                 <chr> "['Big Data', 'Hadoop', 'Hive', 'P~
$ educational_institution_name           <chr> "['The Amity School of Engineering~
$ degree_names                           <chr> "['B.Tech']", "['B.Sc (Maths)', 'M~
$ passing_years                          <chr> "['2019']", "['2015', '2018']", "[~
$ educational_results                    <chr> "['N/A']", "['N/A', 'N/A']", "['N/~
$ result_types                           <chr> "[None]", "['N/A', 'N/A']", "['N/A~
$ major_field_of_studies                 <chr> "['Electronics']", "['Mathematics'~
$ professional_company_names             <chr> "['Coca-COla']", "['BIB Consultanc~
$ company_urls                           <chr> "[None]", "['N/A']", "['N/A']", "[~
$ start_dates                            <chr> "['Nov 2019']", "['Sep 2019']", "[~
$ end_dates                              <chr> "['Till Date']", "['Till Date']", ~
$ related_skils_in_job                   <chr> "[['Big Data']]", "[['Data Analysi~
$ positions                              <chr> "['Big Data Analyst']", "['Busines~
$ locations                              <chr> "['N/A']", "['N/A']", "['N/A']", "~
$ responsibilities                       <chr> "Technical Support\nTroubleshootin~
$ extra_curricular_activity_types        <chr> "", "", "", "", "['Professional Or~
$ extra_curricular_organization_names    <chr> "", "", "", "", "['Ohio Society of~
$ extra_curricular_organization_links    <chr> "", "", "", "", "[None, None, None~
$ role_positions                         <chr> "", "", "", "", "[None, None, None~
$ languages                              <chr> "", "", "", "", "", "", "", "['Spa~
$ proficiency_levels                     <chr> "", "", "", "", "", "", "", "['N/A~
$ certification_providers                <chr> "", "", "", "", "['Ohio Notary Pub~
$ certification_skills                   <chr> "", "", "", "", "[None]", "[None]"~
$ online_links                           <chr> "", "", "", "", "[None]", "[None]"~
$ issue_dates                            <chr> "", "", "", "", "[None]", "['May 2~
$ expiry_dates                           <chr> "", "", "", "", "['February 15, 20~
$ X.job_position_name                    <chr> "Senior Software Engineer", "Machi~
$ educationaL_requirements               <chr> "B.Sc in Computer Science & Engine~
$ experiencere_requirement               <chr> "At least 1 year", "At least 5 yea~
$ age_requirement                        <chr> "", "", "", "Age 22 to 30 years", ~
$ responsibilities.1                     <chr> "Technical Support\nTroubleshootin~
$ skills_required                        <chr> "", "", "Brand Promotion\nCampaign~
$ matched_score                          <dbl> 0.8500000, 0.7500000, 0.4166667, 0~
```